

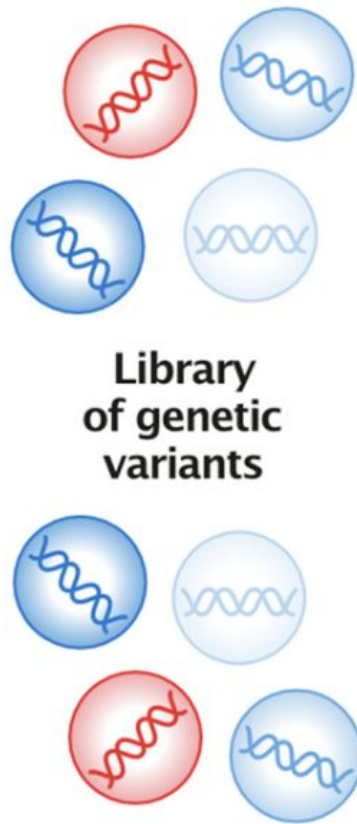
connecting people with science

Dr Alan Rubin

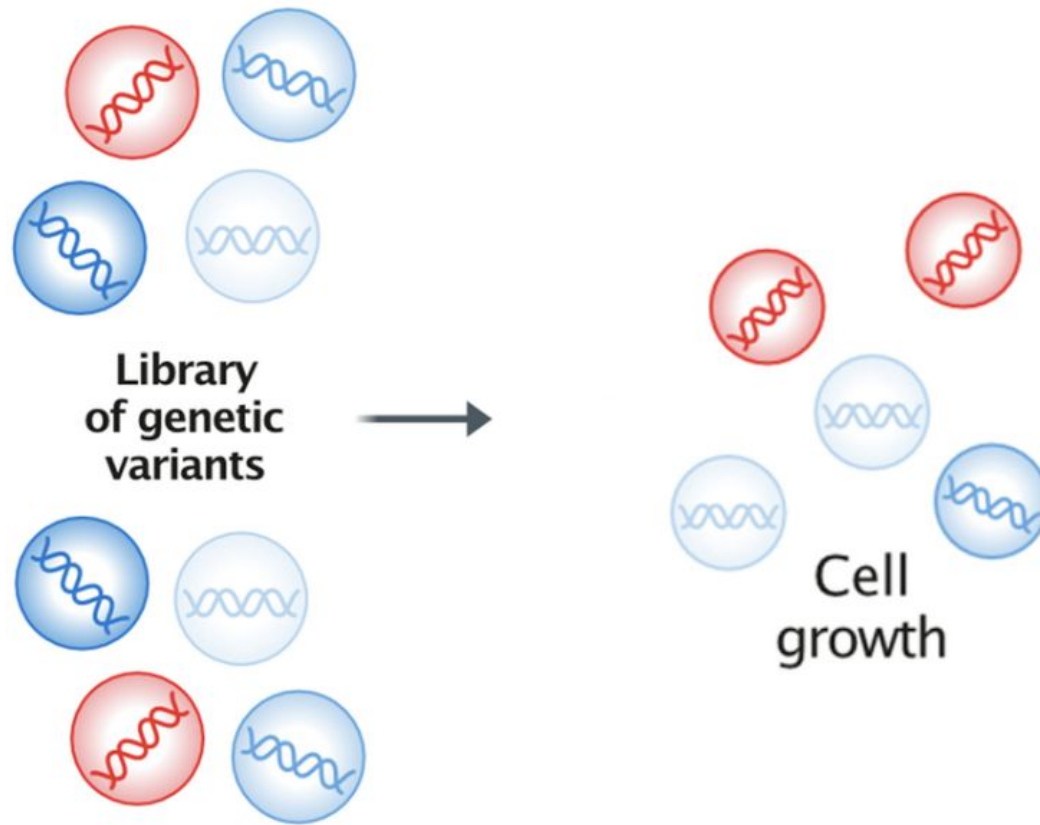
MAVE data analysis II
From read counts to scores

November 2025

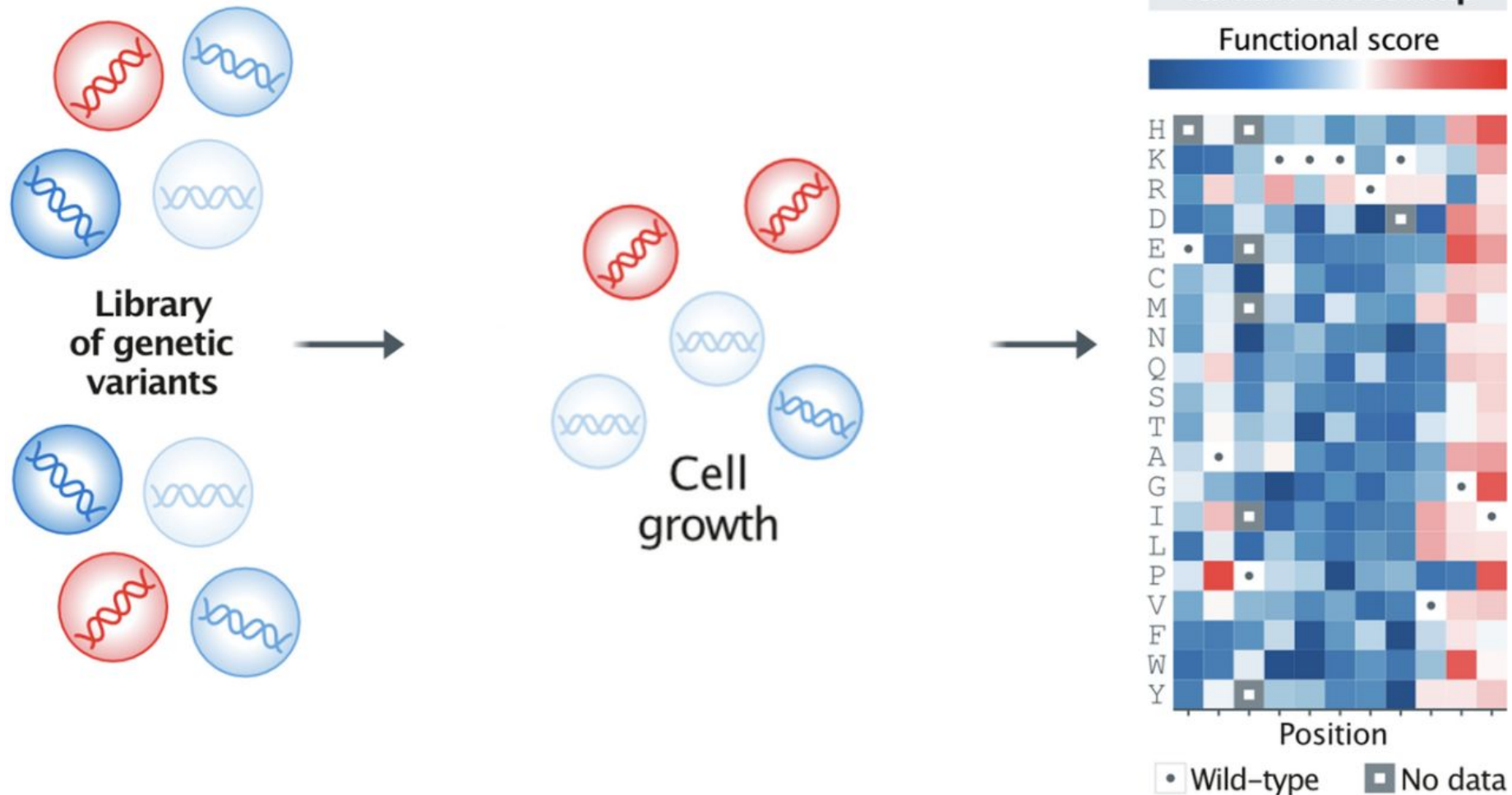
Multiplexed assays of variant effect



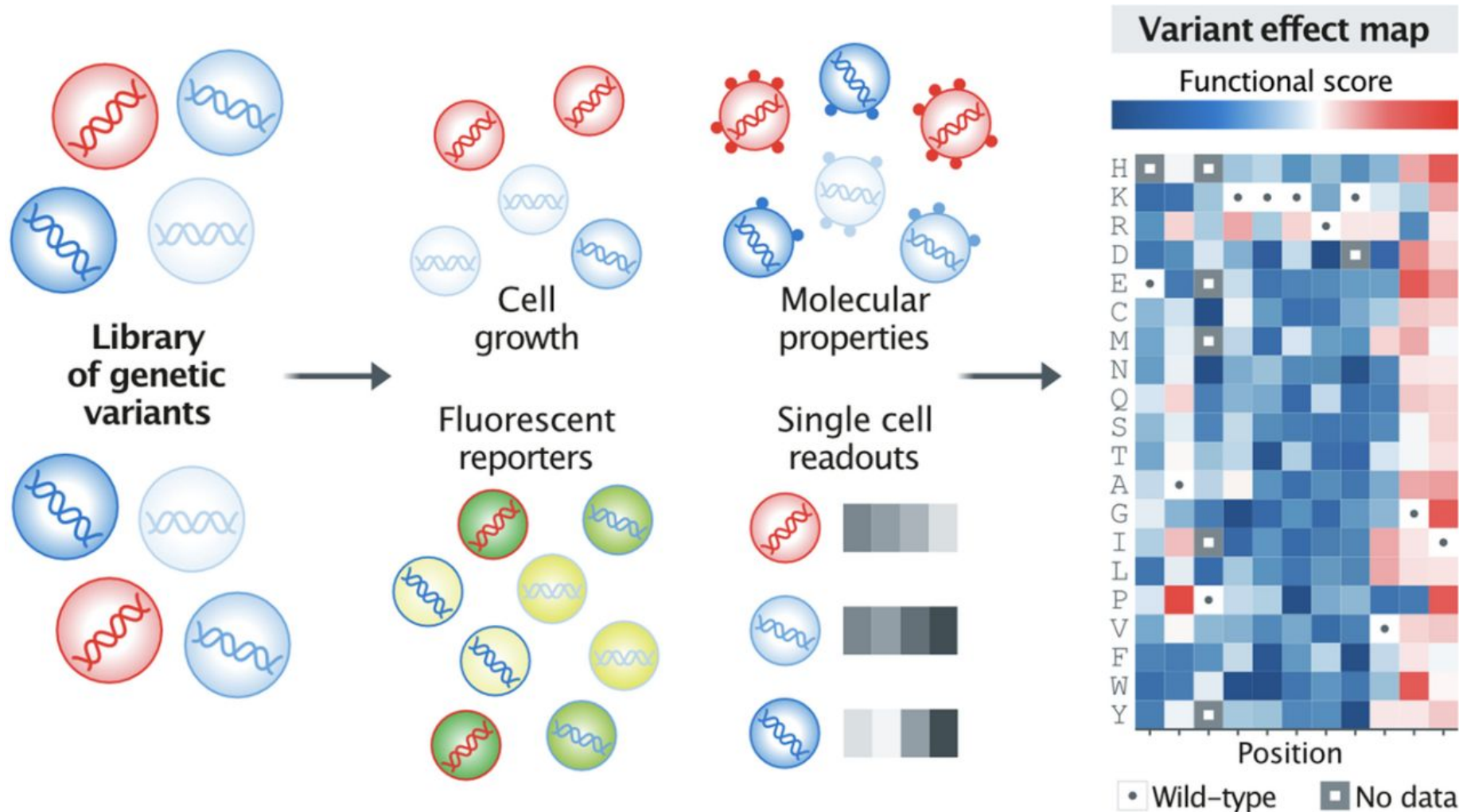
Multiplexed assays of variant effect



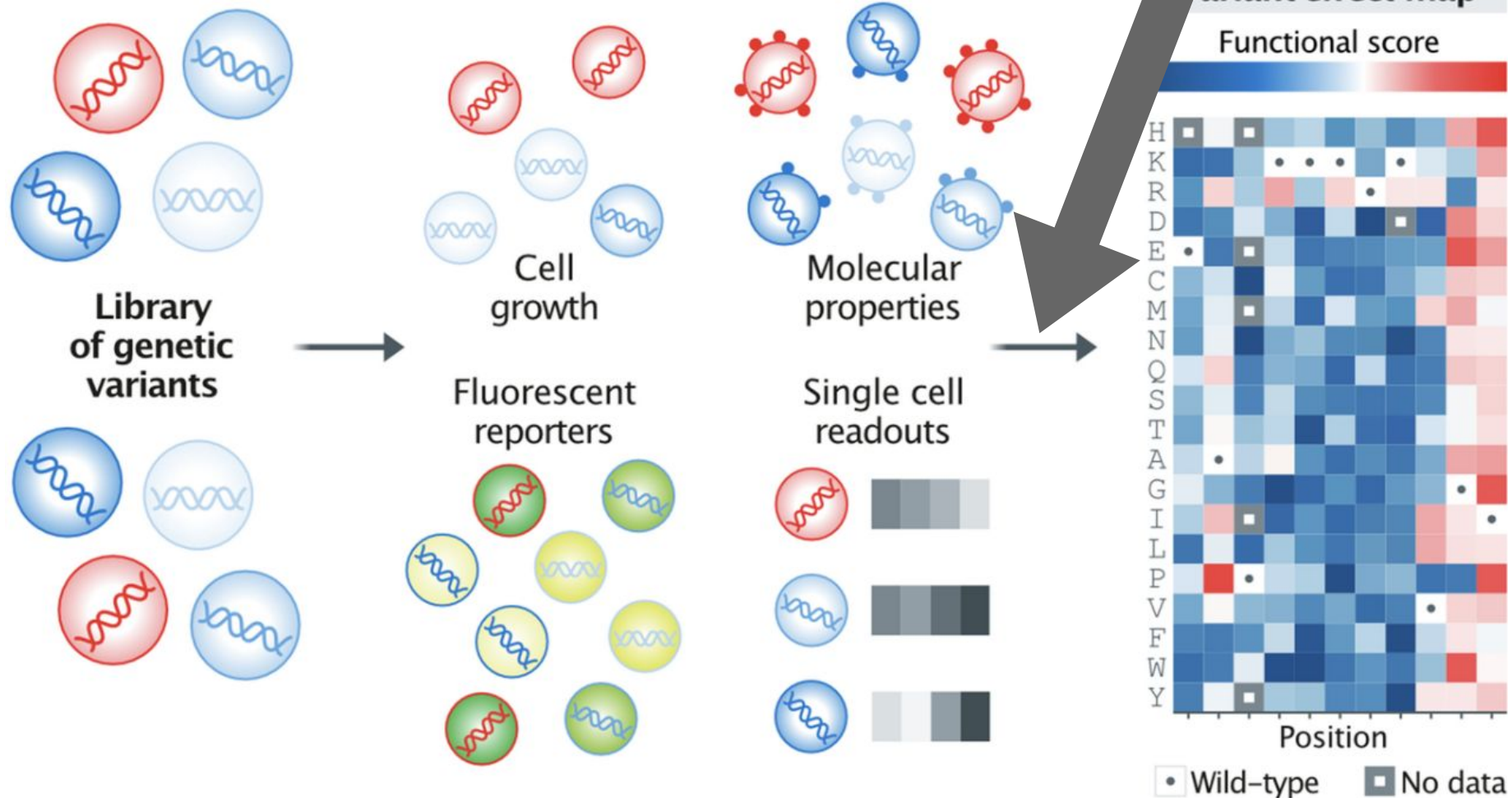
Multiplexed assays of variant effect



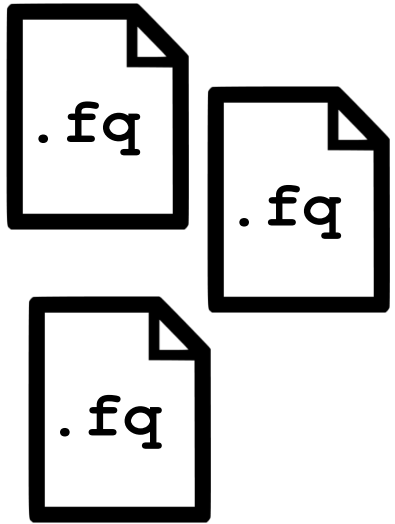
Multiplexed assays of variant effect



Multiplexed assays of variant effect



Calculating scores



<i>Variant</i>	<i>Count 0</i>	<i>Count 1</i>
p.Asp2Glu	10	20
p.Asp2=	20	20
p.Asp2Tyr	10	2



<i>Variant</i>	<i>Score</i>
p.Asp2Glu	1.0
p.Asp2=	0.0
p.Asp2Tyr	-2.3

A simple scoring method

Frequency

$s_v = \log_2$

“Ratio of ratios”

Library size

The diagram illustrates the formula for a simple scoring method, $s_v = \log_2$ applied to a ratio of ratios. The formula is enclosed in a blue rounded rectangle. Inside, the numerator is $\left(\frac{c_{v,\text{output}}}{\sum_i c_{i,\text{output}}} \right)$ and the denominator is $\left(\frac{c_{v,\text{input}}}{\sum_i c_{i,\text{input}}} \right)$. A green rounded rectangle highlights the numerator, with an arrow from the label 'Frequency' pointing to it. An orange rounded rectangle highlights the denominator of the numerator, $\sum_i c_{i,\text{output}}$, with an arrow from the label 'Library size' pointing to it. A blue arrow points from the label '“Ratio of ratios”' to the blue rounded rectangle. The label $s_v = \log_2$ is positioned to the left of the formula.

$$s_v = \log_2 \left(\frac{\left(\frac{c_{v,\text{output}}}{\sum_i c_{i,\text{output}}} \right)}{\left(\frac{c_{v,\text{input}}}{\sum_i c_{i,\text{input}}} \right)} \right)$$

Additional concerns for MAVE scoring

- Count filtering
- Experiments with more than two time points (regression)
- Bin-based experimental designs
- Other types of ratios (e.g. wild-type normalisation)
- Combining replicate measurements
- Quantifying uncertainty

Review Article



molecular
systems
biology

Variant scoring tools for deep mutational scanning

Hasan Çubuk ^{1,4}, Xinyi Jin ^{2,3,4}, Belinda Phipson ^{2,3}, Joseph A Marsh ¹✉ & Alan F Rubin ^{2,3}✉

Molecular Systems Biology Volume 21 | Issue 10 | October 2025 | 1293–1305

Learning objectives

- Learn what MAVE data looks at each stage of the process
- Implement your own simple approach and compare it to more sophisticated methods
- Develop an understanding of how analysis decisions affect the results of the analysis

Lesson outline

- Brief background lecture
- Implementing your own simple scoring approach for analysing SGE data
- Data exploration and experimentation

Jupyter notebook link

WCSCourses / MAVE_2025 Public

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 afrubin

add notebook for mave data analysis 2

e71a22d · 3 days ago

Name	Last commit message	Last commit date
 ..		
 README.md	add notebook for mave data analysis 2	3 days ago
 basic_MAVE_scoring.ipynb	add notebook for mave data analysis 2	3 days ago

A simple scoring method

$$s_v = \log_2 \left(\frac{\left(\frac{c_{v,\text{output}}}{\sum_i c_{i,\text{output}}} \right)}{\left(\frac{c_{v,\text{input}}}{\sum_i c_{i,\text{input}}} \right)} \right)$$

